

Typing pattern matching

We use possible/surely accepted types to type case expressions:

Consider $\text{case } e \text{ do } p_1 g_1 \rightarrow e_1 ; \dots ; p_n g_n \rightarrow e_n$ with $e : t_0$

the type t_i of all values that may be captured by the i -th branch is

$$t_i = (t_0 \setminus \bigvee_{j < i} \{p_j g_j\}) \wedge \{p_i g_i\}$$

In words: the expression e_i will process values:

- generated by e (i.e., in t_0)
- **minus** those surely captured by the previous branches (i.e., in $\bigvee_{j < i} \{p_j g_j\}$)
- **and** that may match p_i **when** g_i (i.e., in $\{p_i g_i\}$)

$$\text{(case)} \quad \frac{\Gamma \vdash e : t_0 \quad (\forall i < n \quad \Gamma, (t_i/p_i) \vdash e_i : s_i)}{\Gamma \vdash \text{case } e \text{ do } \overline{p_i g_i} \rightarrow e_{i < n} : \bigvee_{\{i | t_i \neq \emptyset\}} s_i}$$

use only the branches that may be selected (redundancy)

derive the types of the variables in p_i from t_i

exhaustiveness condition

$$t_0 \leq \bigvee_{i \leq n} \{p_i g_i\}$$